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Enlarging the Diegetic Space: Uses of the Multi-channel Soundtrack in Cinematic Narrative

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Abstract

Throughout its evolution, cinema sound has changed the way we watch and understand films. Initially sound carried little narrative information. Today's multi-channel soundtrack, on the other hand, is able to enlarge the diegetic space and immerse the audience in it. However, it could be argued that the multi-channel soundtrack's potential as a narrative tool has yet to be fully explored. This article reports on our research into the possibilities and limitations of the multi-channel soundtrack as a carrier of large sections of narrative information. We identify three areas of interest: spatialisation of sound sources; surround ambiences with a narrative relevance; and separation of sound streams. A short film has been created exploring these topics and feedback has been elicited from a small audience. We conclude that this potential could be fully unlocked if considerations regarding the surround soundtrack are taken into account in the early stages of production.

Keywords

multi-channel sound
surround sound
cinematic narrative
spatialisation of sound sources

Introduction

It is not surprising that sound has formed a fundamental element in the cinematic experience since its integration with the visual action. Even the so-called silent movies were never truly silent, because since the dawn of cinema, music was provided for films and performed while they were shown (Chion 1999: 8). Over the years sound has secured its place in the cinematic context: a place that, according to many, is as significant as that of visuals (Kerner 1989) (Szaloky 2002: 112) (Chion on Sinclair 2003: 20) (Sinclair 2003: 17).

With the technological advances that took place in the previous century, sound took large steps forward. Among these steps was the transition from mono to two-channel stereo sound and later to surround sound (Holman 2000: 13–24). Although at the beginning of the surround sound era many might have believed that an entirely new audio world with unlimited possibilities had been discovered, we believe that the largest part of surround's potential has yet to be explored. This is mainly because although the multi-channel (5.1 surround) audio system is now the standard in cinema, it is often used as an extension of traditional stereo rather than a 3D audio canvas capable of conveying crucial narrative information (Holman 2000: 11).

This article explores the possibilities of exploiting and using more extensively the potential of today's multi-channel systems for enhancing cinematic narration. Although these topics and ideas are not new, it is believed, as stated above, that the potential of surround sound has yet to be fully uncovered and, consequently, that there is much scope for experimentation and exploration in this context.

Additionally, it was decided that this work should also include the creation of a short film in which the ideas and the suggestions made during the research could be integrated, tested and evaluated. As a result, a short film was produced, and a series of viewings involving ten participants was carried out, in order to provide an initial response regarding the effectiveness of such surround sound techniques.

3D sound and the expansion of the sound design scope

One of the most significant recent changes in audio technology has been the transition to multi-channel (surround) systems. Although today's technological advances in the field have created the potential for a great sound revolution in cinema and entertainment in general (Bridget 2005), many 'have simply ignored the amazing mutation brought on by the standardization of Dolby sound' (Chion 1994: 152). The reasons for this are various and multifaceted, yet it can be claimed that the visual component is still considered by many to be the most important part of the cinematic experience, and as a result, sound is still effectively treated as a secondary and mainly two-dimensional complement which follows the visual action. Perhaps some practitioners are prevented from taking the required risks by the implications, and huge technical and aesthetic changes, that a move towards a cinema that better utilizes surround sound would entail (Kerins 2006). As Leider stresses:

> Working in surround instantiates an immense, additional set of aesthetic considerations and choices. These include, but are of course not limited to, an expanded questioning of the purpose of the loudspeaker during playback in the mix; the portrayal, creation, or deconstruction of acoustic reality; questions of auditory perspective; and the overriding notion of space as a compositional and mixing parameter.
>
> (Leider 2004: 277–278)

Despite these risks, many people nowadays are indeed interested in finding new ways to exploit existing surround sound systems, and significant progress has been made over the last few decades in that respect. Moreover, although 'surprisingly few film scholars have taken much notice of digital surround sound' (Kerins 2006: 41), surround has been used in extensive and creative ways as a narrative tool in many major animation and movie productions. Examples are: *Toy Story* (1995) where 'the filmmakers have expertly exploited DSS's multi-channel capabilities to create enveloping, immersive environments that also forward their stories' (Kerins 2006: 44); *Finding Nemo* (2003) where 'sounds are panned within the five-channel sonic space to create the feeling of a complete world surrounding us' (Kerins 2006: 45); and *Saving Private Ryan* (1998) where 'the soundtrack forms a consistent frame of reference, which helps us

keep a handle on spatial relations, regardless of what the image track does' (Kerins 2006: 52).

From a technical point of view, surround sound has mainly influenced the way visuals are edited. Chion mentions that surround soundscapes 'logically had the effect of undermining the narrative importance of the long shot' (Chion 1994: 150) and they 'progressively modified the structure of editing and scene construction' (Chion 1994: 150). In particular, information about the context of a scene can be portrayed through the surround soundscape while the images can focus on more intimate details.

However, the influence of surround sound on editing is still to be explored by many film-makers. If one closely observes today's cinematic creations it becomes apparent, despite the advances and the changes mentioned above, that audio often continues to function mainly as a complement to the visuals.

Therefore, although surround sound has already brought significant changes to the ways film-makers approach cinematic creation, it can be claimed that, in order to take full advantage of its existing potential, a fundamentally different cinematic production process is required. Such a process would consider 3D sound as a basic directing tool rather than a mere complement to the pre-determined and, for this reason, inflexible visual edits.

An overview of today's cinematic surround sound

Before discussing possible ways in which 3D sound can affect cinematic narrative, it is useful briefly to look at some facts related to the current state of 3D sound systems. Many attempts have been made towards the accurate representation of 3D soundscapes in the twentieth century, and these have resulted in the creation of several different 3D sound theories and technologies (Rumsey 2001: 10–20) such as Quadraphonic, Ambisonics, Wavefield Synthesis (IRCAM 2004) (Hofman 2007) (Daniel 2000) (Ambisonic.net 2007). Most of these and other similar systems, which are not specifically designed with cinematic sound in mind, aim at representing a 3D soundscape in a realistic way; they give the same importance to the movement and placement of sound in all directions, often using a speaker array configuration that covers equally all areas of space around the listener. It appears that this realistic spatial representation of sound events was not convenient for cinematic use for various reasons. This was mainly due to the fact that the visual cinematic action occurs in a two-dimensional space (the screen) in front of the audience. Additional factors were considerations of: the auditory perception of sounds in space, which is not the same in all directions (Holman 2007: 180); the perception of multiple simultaneous audio streams, which can be improved when audio streams are spatially separated (Bregman 1990: 642–649); the audio-visual relationship (Sinclair 2003: 21); and, in particular, sound magnetisation by the screen action (Chion 1994: 69–70) (Cancellaro 2006: 167). Sound magnetisation refers to the phenomenon by which we tend to accept that the sound of characters and objects on the screen come from their visual position, even when the sounds are actually emanating from a single central speaker. Today's cinematic sound, although it includes surround elements, is assumed just to support the action presented visually on the two-dimensional screen and, to an extent, suggest some spatial movement. This need for a

1. There are also non-screen oriented surround systems for music spatialisation in use, by which speakers are evenly distributed around the audience and sound information is equally important in each direction.

screen-oriented surround sound system eventually led to the creation, and commercial standardisation, of 5.1 and 6.1 systems such as Dolby Digital and THX. In these systems an array of speakers primarily cover the screen area, leaving only a few speakers to cover the sides and the back.[1] These are arguably the most popular and broadly used cinematic surround sound systems today (Silva 2008) (Rumsey 2001).

Using such systems, and after experimentation with different sound spatialisation techniques, contemporary sound designers commonly employ what have come to be regarded as 'standard' techniques of surround sound use. Generally, the different elements or stems (Beauchamp 2005: 29) of the soundtrack, such as dialogue, music and sound effects, are placed in space according to conventions related to the visual action (Beauchamp 2005) (Cancellaro 2006) (Wyatt 2005). Typically, dialogue is placed in the centre front speaker and music in the front right and left speakers (in stereo); this leaves the surround speakers (and the subwoofer) for sound effects and atmospheres (Wyatt 2005: 206) (Beauchamp 2005: 145–149) (Holman 2000: 112). This way, separation between the elements of the soundtrack becomes clearer. However, the strict application of the above spatialisation techniques could clearly reduce the freedom of placing the sound samples in a 3D soundscape: something that consequently makes the creation of an immersive 3D soundscape difficult to achieve.

Areas of cinematic narrative in which surround sound can be applied more extensively

Among many others, we believe that some of the areas of cinematic narrative that can be greatly influenced by a more effective use of 3D sound are:

- the spatialisation of moving and static sound sources;
- the creation of surround ambient soundscapes with narrative character;
- the separation of different sound elements of the cinematic action, such as internal and external sounds (Chion 1994: 76) and simultaneous sound streams.

Spatialising sound sources

Firstly, it is useful to stress the differences between the notions of sound and sound source (Chion 1994: 79) (Rumsey 2001: 1–6). According to Szaloky:

> Sound is seldom, if ever, defined by its own aural characteristics but rather is understood as an attribute, an acoustic accompaniment, of the visible source that emits it. This also explains the term off-screen sound, 'which in actuality refers to the spatial location of the source of the sound rather than the sound itself'.
>
> (Szaloky 2002: 112)

Chion also mentions that 'spatially speaking, the sound and its source are two different entities' (Chion 1994: 79). In the surround context, the actual task is to find a way to convince the audience that an off-screen sound source is located somewhere in the 3D space surrounding the audience – by means of sounds that suggest position, distance and nature (Cancellaro 2006: 167). Obviously this could be achieved by placing a

sound that is supposed to be emitted by the relevant sound source at a specific point in the 3D soundscape. A related problem, however, is the tendency of sound to be *magnetised* by the visual action (Chion 1994: 69). This is true 'only if the sound itself maintains a basic spatial stability. If it [the off-screen sound] constantly moves back and forth among loudspeakers, the image will have a harder time absorbing it' (Chion 1994: 70). Therefore it is worthwhile to consider the possibility, when appropriate, of using more extensive sound movements in order to point the audience's attention to the off-screen surround audio.

Relevant to the above, mentioned by Chion (1994: 84), is the idea of expanding the cinematic action space in an imaginative arena that is bigger and broader than the picture; this can be done by means of extending the movement of on-screen objects (sound sources) beyond the screen (and these sound movements continued off-screen). This approach, known as the 'in-the-wings effect', was tried in the past and inevitably introduced continuity and editing problems, as the sound events did not follow the visual action temporally (Chion 1994: 83–84). Examples of the above could be 'the footsteps of a character approaching or leaving, or the engine of a car that has just gone off-screen', in order to make the audience believe that 'the audio-visual space is being extended beyond the borders of the screen' (Chion 1994: 83–84). This is often 'disconcerting to our normal sense of spectatorship' (Chion 1994: 83–84). 'Slowly, this practice has been dropped ... [as] it created a nagging problem by violating the conventions of continuity editing and making sound matching problematic' (Chion 1994: 84). However, if the design of the soundtrack is part of the scripting and the pre-production process, rather than a complement to the visuals, then both the production and the editing processes could be of a very different nature, and the consequent solution of the above issue could result in enhanced cinematic narration. As Chion suggests, 'maybe it [the 'in-the-wings' effect approach] could have gained more permanent admittance into film practice had it been systematised along with some partial adjustments in editing conventions', and 'perhaps it was a mistake to have given it up so quickly' (Chion 1994: 84).

The short movie produced in the context of this research addressed some of these issues by significant use of off-screen sound, as well as by reducing to a minimum the presentation of sound sources simultaneously with visuals. This reduced both the problems of sound magnetisation to the screen and the 'in-the-wings' effect. Moreover, the sound design for this short film attempted to treat the surround sound system (in this case a 5.1 speaker system) as a palette for 3D sound design, rather than a way to separate different soundtrack stems conventionally.

Creation of surround ambient soundscapes

Sonnenschein's claim that 'the integration of [surround] sound and image can create a new perception of space' (Sonnenschein 2001: 161) leads us to another important issue: surround sound can be used not only in order to describe spatially sound sources and their movements, but can also be used as a convenient way to create 3D ambient soundscapes, which immerse the audience and create the required envelopment and spacious elements (Holman 2000: 118, 215–216).

Chion uses the term 'superfield' to describe these 3D cinematic soundscapes.

> I call *superfield* the space created, in multitrack films, by ambient natural sounds, city noises, music, and all sorts of rustlings that surround the visual space and that can issue from loudspeakers outside the physical boundaries of the screen.
>
> (Chion 1994: 150)

Although 'today's multipresent sound has insidiously dispossessed the image of certain functions, for example, the function of structuring space' (Chion 1994: 144), we can question to what extent we are using the potential of surround sound as a tool to create ambiences with narrative importance.

Before we examine this issue in more detail it is useful briefly to present the notion of active and passive sounds. An active sound is a sound that 'raises questions – What is it? What is happening?' (Chion 1994: 85). On the other hand 'passive sound creates an atmosphere that envelopes and stabilises the image without inspiring us to anticipate its source' (Chion 1994: 85).

Many believe that 3D ambient soundscapes, containing such passive and active off-screen sound elements (Chion 1994: 84) can be recreated convincingly using existing surround sound technologies. Within these ambient soundscapes one could place any kind of distinctive realistic or cinematic sounds that could, as mentioned previously, not only create or enhance the overall ambience, but also simultaneously act as static or moving sound elements emanating from distinct sound sources. As Barrett claims: 'when you consider very tiny sounds building up an impression of space, then you can begin to imply a space within which these sounds should live' (Barrett 2007: 13). It is suggested that in this way the ambience itself becomes a surrounding active organism that can include as much or as little audio information as the cinematic narration requires.

Also in this case, if these soundscapes and their use were designed during the pre-production process and had a clear and active narrative role, we believe that they could then make better use of even the existing surround sound systems. They could be easily transformed from passive ambiences, which are designed merely to enhance the atmosphere and information conveyed through the visuals, to fully active and immersive worlds containing very important narrative information.

Surround ambiences with both passive and active sounds, and high narrative importance, were created for the short film produced for this research.

Using surround sound for the separation of different sound elements

Although the spatialisation of sound sources and ambient soundscapes constitutes one of the main areas that can effectively utilize the possibilities of 3D sound, spatialisation is by no means the only possible way to use surround sound within a cinematic context. The positioning and movement of sound elements in space can increase the ability of the listener to distinguish different sounds and allow him/her to follow multiple sound streams

more easily (Moore 1997: 249) (Holman 2000: 117). If sound spatialisation facilitates the separate perception of many different (also simultaneous) sound elements, this characteristic could be exploited in order to convey useful narrative information in an innovative way. One potential of the use of this idea might be the separation of internal and external sounds (Chion 1994: 76) or the separation of the narrator's voice from diegetic sounds and voices (Cancellaro 2006: 164). It is suggested that as far as the listener is capable of following at least two sound streams simultaneously, the left and right surround speakers can be used for the internal voice of a character or for the narrator's voice; this leaves the front speakers free to carry all the other information from external or diegetic sources.

Expanding this, one can imagine that a 3D sound system could provide the means to separate and place into space as many simultaneous sound streams as required by the story: assuming that these would be kept, to a reasonable extent, spatially separate, in order to allow the audience mentally to identify them. In reality, there will be a limit to how many streams of sound can be perceived simultaneously and this limit is highly dependent on the type of sound material constituting the sound streams, its frequency range, whether it is speech, noise or music, etc. It is suggested that this approach might be useful in situations where the camera conveys the viewpoint of one of the cinematic characters, known as point of view camera (POV). In this case, the positioning of objects, events and characters around the camera can be represented by distinct and possibly simultaneous audio streams. Examples of this could be: the character has hallucinations represented by voices and sounds; or the character and another person are arguing and thus talking at the same time.

Two explicit examples of sound separation through spatialisation were included in the short movie created for this work. In one case, this took the form of the separation between the main character's thoughts (rear speakers) and his actual voice (front speakers). In order to evaluate the separation due to the spatialisation between these two streams, no additional processing was applied to either: thus the only clue for the audience would be the sound streams' position in space.

In the second case, four simultaneous sound streams of similar audio material, representing off-screen sounds that provided a flashback for the main character of the movie, were spatially separated and introduced into the film in order to provide a way of testing spatial separation of simultaneous sound streams.

The short film produced for this research

Various factors were considered when creating the narrative for this short film, so that it could provide the necessary context for the required surround sound work. It was decided that the story should be flexible enough to allow rapid transitions between many different situations. For this reason, the idea of presenting the chronicle of a car accident through the confused thoughts and memories of one of the survivors was employed. In such a narrative, realistic, fictional and possibly exaggerated elements could be justified.

The story starts with the main character (Frank) in an operating theatre just before surgery. As Frank slowly falls into anaesthesia, he experiences memories related to the car accident. These memories, presented to the

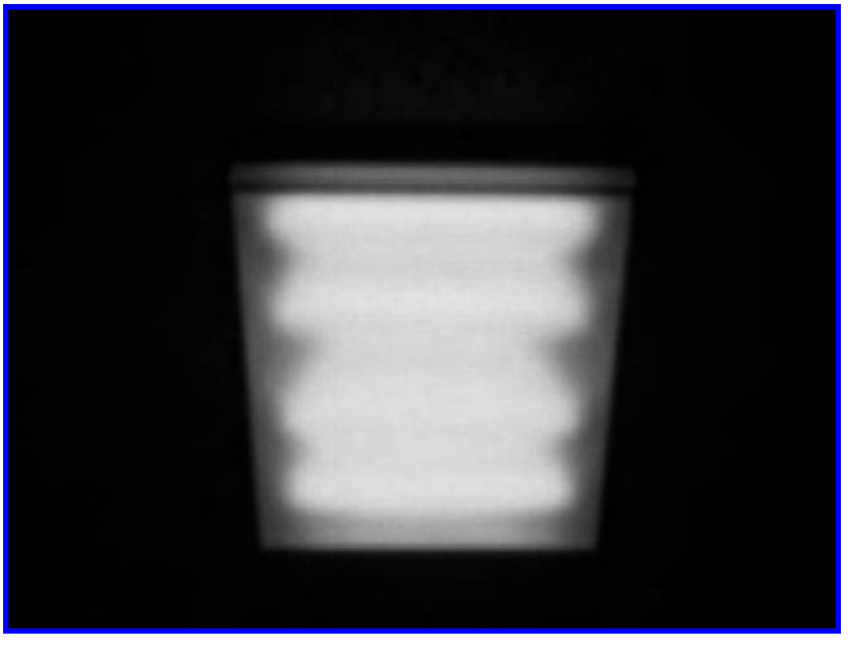

Figure 1: The surgery room.

audience in random chronological order, chronicle the accident piece by piece primarily through the audio; since the visuals, representing the main character's point of view, are intentionally void of narrative clues. These images are relatively static and have undergone minimal editing. Figures 1 to 6 show the six scenes presented in the film: the surgery room; the

Figure 2: The accident.

Figure 3: In the car before the accident.

accident; in the car before the accident; Frank's house; Frank's office building; and the café where Frank is talking to his colleagues.

After the parts of the story have been presented through Frank's memories, the last scene takes the audience back to the surgery room where he is slowly recovering.

Figure 4: Frank's house.

Figure 5: Frank's office building.

A great variety of scenes can be created for such a flexible story; both realistic scenes where the character is at the hospital, and fictional scenes where the memories of the car accident are distorted by the character's emotional and/or mental state. This flexible structure provided a very good basis for experimentation. The soundtrack was required to provide

Figure 6: A coffee with colleagues.

effective transitions between the realistic/real-time hospital scenes and the dreamy, metaphoric ambiences created for the scenes where Frank recalls fragments of the accident and his life.

The dreamy flashback sequence also provided the potentially more abstract environment, in which the effectiveness of the experimentation with surround sound could be more freely produced and successfully evaluated. One reason for this is that in a sequence which combines realism with non-realism, such as the one described above, surround sound and spatialisation techniques could be used to clarify and underline crucial narrative information that could otherwise be missed.

Evaluation of how well the surround soundtrack carries narrative information in the short film

An audience of ten people was asked to view the short film in order to evaluate how well the surround soundtrack carried important narrative information. The aim of this test was to gather external feedback from an audience, which was unaware of the focus of the study. It also served as an indication of the success (or otherwise) of the techniques employed. By no means is this test intended to be a scientific experiment on human perception of sound spatialisation in a cinematic context. Such an experiment would require a highly controlled design of the stimuli and a much larger sample of participants to produce statistically significant results.

This study could be considered, instead, as an initial exploratory study useful to evaluate and stimulate constructive new comments on the research ideas it explores.

The test was divided into two parts: the first evaluated how much narrative information was carried by the soundtrack; and the second how effective the surround sound techniques used were in making the narrative information in the soundtrack clear to the audience.

The first part of the test involved viewing the movie initially without the soundtrack (as a silent movie), and then with the soundtrack. The subjects were then asked to give brief overviews of the story as they understood it initially, relying only on the visuals, and then by being assisted by the soundtrack as well. By comparing the changes that the soundtrack introduced to their perception, it was hoped that useful conclusions regarding the importance of the soundtrack as carrier of narrative information could be drawn.

After the viewings, for the second part of the test, the subjects were asked to fill in a questionnaire (see Appendix), which was intended to measure the effectiveness of the techniques used. The questionnaire consisted of a series of closed questions focused on the perception of specific elements of the soundtrack. Example of questions were 'How many people can be identified in the accident location?', 'Where are they located?', etc. In order to complete the questionnaire, the participants were asked to watch the short film again, this time scene by scene, *after* they read the questions relevant to each scene. In this way the subjects were allowed to focus on particular elements and information conveyed by the soundtrack, elements that in normal *watching-film* mode would be absorbed by the audience quite unconsciously. This allowed us to verify whether these

elements were identified correctly and deduce if their presentation in the surround soundtrack was sufficiently clear.

Summary of observations from the questionnaire

Some useful observations were made by analysing the small audience's answers to the test.

Firstly, as one would probably expect, a marked difference was identified, in terms of detail, between the summaries of the story produced after the silent version's viewing as compared to the audio version. Most of the participants could guess the story plot at a basic level after viewing the silent version, but they were unable to identify crucial narrative elements, such as the exact details of events, the resolution of the story or the main characters and their relationships. Considering that the movie was undeniably sound-oriented, this is not surprising. However, these findings strongly reinforce Chion's claim that 'sound still has the role of showing us what it wants us to see in the image' (Chion 1994: 144). This confirms the power of sound as a cinematic element; a power and potential that can be exploited to its maximum when a great deal of thought is given to the surround soundtrack during the pre-production and production stages.

In this short film many important narrative elements were conveyed through the positioning of sound elements within the 3D soundscape. The positions and movements of objects and characters in the scenes were portrayed through their positions in the surround field only, without the use of visuals or editing (the visuals, as mentioned previously, carried very little narrative information and were deliberately quite static; and editing was reduced only to the transitions between different scenes). Figures 7 and 8 show two examples of sound positioning in the surgery and car scenes.

Generally the audience identified the rough positions and movements of objects and characters that were portrayed, in the mix, principally through the surround panning of the relative sounds.

Surround ambient soundscapes with passive and active sound elements were created for each scene. These soundscapes helped to clarify the characteristics of each different location (the surgery room, the car, the café, etc.). The passive elements of the soundscape clarified and confirmed information that was also partially presented by the visuals, while active sound elements carried the narrative forward.

In various instances it was observed that while the audience could perceive all the sounds present in the soundscape, they could not clearly separate the passive ambience elements from the active sounds (which carried the intended narrative information). This suggests the possibility that in order to make the narrative clearer, more sophisticated techniques would be needed to direct the audience's attention to the active sounds.

Spatial separation in the surround field provided quite a successful way to separate Frank's thoughts from his actual voice. The majority of the audience understood the difference between the two speech streams; only a few seemed to be a little confused. Such a spatialisation choice clearly influences the construction of shots and the picture editing, and this choice needs to be made during the pre-production and production stages.

Although spatial separation worked well for the presentation of Frank's internal and external voices, attempts that were made to use multiple,

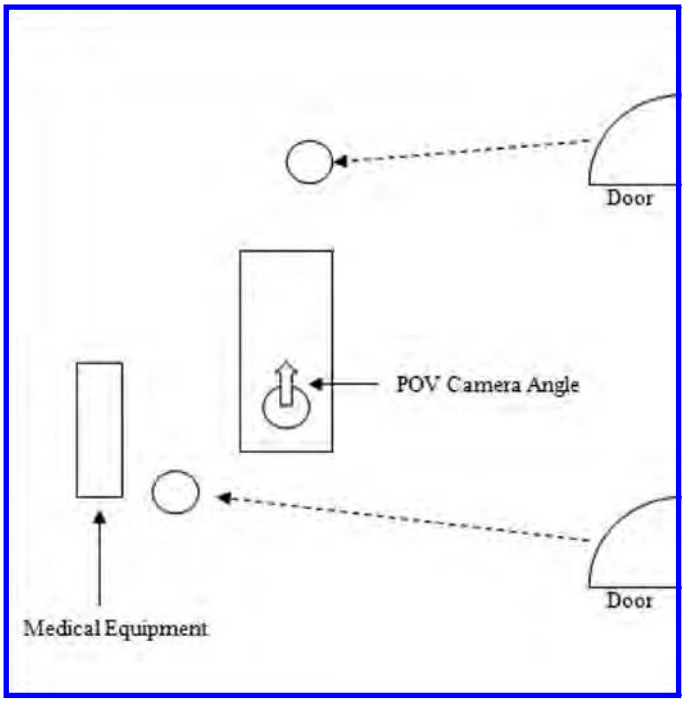

Figure 7: Positions of sounds in surgery room.

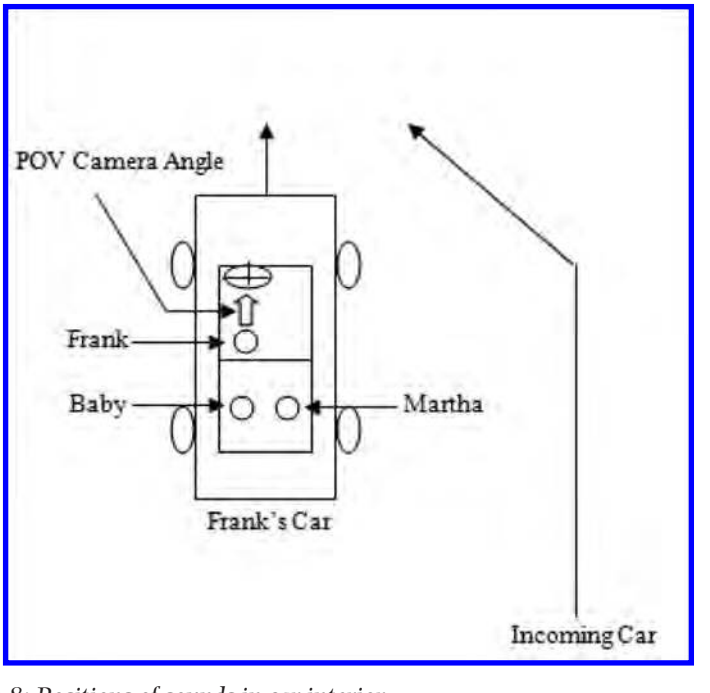

Figure 8: Positions of sounds in car interior.

simultaneous, spatialised sound streams (similar streams in terms of audio material and narrative significance) did not have the expected result. This spatial separation was created in the scene where Frank remembers his house and family. The image shows the house, while the soundtrack presents four simultaneous audio streams of familiar sounds in Frank's life: the dog barking; the baby crying; the daughter saying hello; and the birds singing. The main reason for the unclear separation of the sound streams is believed to be the short durations of the streams, which did not give the audience sufficient time to focus selectively on the simultaneous sound events. Sound stream separation through spatialisation is a complex field that can form the basis of a separate study in its own right. It is suggested that a more detailed study, focused particularly on this specific topic, could provide more valuable information regarding the creative use of simultaneous streams in surround soundtracks.

Overall these results are encouraging and suggest that further exploration of the three approaches to surround sound design identified in this research can lead to interesting changes in the relationships between narrative information, shot construction, visual editing and the soundtrack. A story with a more complex storyline, in which the positions and movements of objects are more important for the understanding of the plot, would allow more scope for this type of experimentation and could be used for future work.

Conclusions

Although surround sound is used extensively in contemporary cinema, we believe that the potential of this technology has not yet been fully explored. We have offered some suggestions regarding the possible ways in which surround sound could be used to enhance narrative cinema. The test suggested that the, albeit small, audience seemed to gain an enhanced perceptual representation of the cinematic space and action when surround sound techniques were extensively and specifically used for story-related information.

We believe that the sound design process should be regarded not merely as a post-production procedure, but as a fundamental element throughout the production of any film. The extensive use of such techniques would not only redefine the need for specific visual elements, but also change the approach taken during their production and editing. Were this to be done as a matter of course, cinema would begin to take full advantage of the immense potential of surround sound.

N.B. A 2.0 stereo version of this film is viewable with the e-version of this journal.

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Appendix: The Questionnaire

Part 1: Evaluation of the narrative significance of the soundtrack

In this part of the test you are going to watch a short movie twice. The first viewing will be with no soundtrack and the second with it. Please answer the question below as appropriate.

1. Please summarize the story of the short movie (silent version):
2. Please summarize the story of the short movie (sound version):

Part 2: Evaluation of the narrative effectiveness of the short movie

In this part of the test you will be answering questions regarding specific elements of the short movie. You are going to view different sections of the movie separately so please read carefully and answer every question before you move to the next.

Scene 1 – Surgery Room

1. Where are the medical devices located?
2. How many doors are there in the room? Please describe the position of the door/doors in space.
3. How many persons are there in the room? Please describe each person's position in space.

Scene 2 – Accident Spot

1. How many people can be identified in the accident location? Where are they located?
2. How many vehicles can be identified? Where are they located?

Scene 3 – Car Interior

1. How many people can be identified in the car? Please describe their exact positions.
2. How many cars are involved in the accident? Which directions do they come from or are located in?

Scene 4 – House and Office

1. Frank claims that 'nothing belongs to us, not even the things we consider ours'. Please name as many of the things he talks about as possible.
2. Did Frank and his wife work in the same place? Please justify your answer.

Scene 5 – Bar

1. Where are Frank's friends located?

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Transnational Cinemas

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Aims and Scope

Transnational Cinemas has emerged in response to a shift in global film cultures and how we understand them. Dynamic new industrial and textual practices are being established throughout the world and the academic community is responding. Our journal aims to break down traditional geographical divisions and welcomes submissions that reflect the changing nature of global filmmaking.

Call for Papers

Transnational Cinemas covers a vast and diverse range of film related subjects. It provides a new and exciting forum for disseminating research. The editors are seeking articles, interviews, visual essays, reports on film festivals and conferences. Articles should be up to 6,000 words in length and should be written in English, with all quotations translated.

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